COMP 560: Artificial Intelligence

Learning Rate Optimization using Q-Learning in SGD

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What is the need for a novel approach?

- The performance of gradient-based optimization methods in training models is heavily influenced by the topology of the loss function associated with the model.
- Saddle points are points characterized by having a gradient that is close to zero in all dimensions
- This stalling can mislead the optimization process into prematurely declaring the saddle point to be the minimum.
- The picture shows how gradient descent can stop at a point that is far away from the minimum due to the near zero gradient in all direction.

Benchmark Adaptive Algorithms

Fixed LR

Keep the learning rate constant throughout the training process – difficult to find the right value

Adam

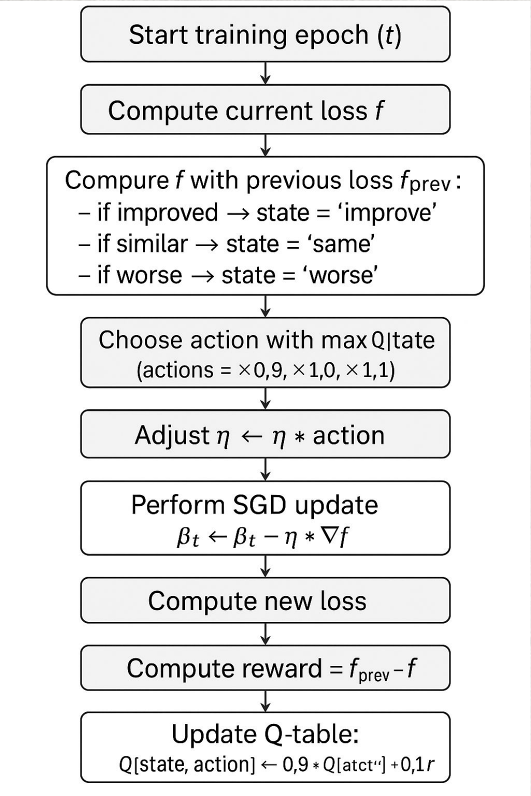
Combines momentum and adaptive learning rates using moving averages of gradients and their squares

AdaGrad

Adapts the learning rate for each parameter by scaling it inversely proportional to the square root of the sum of all historical squared gradients.

SGD Nesterov

Improves standard momentum by computing gradients at a lookahead point, giving a more accurate update direction for faster convergence.


Start training epoch (t)
Compute current loss f
Compure f with previous loss f_{prev}:
– if improved → state = ‘improve’
– if similar → state = ‘same’
– if worse → state = ‘worse’
Choose action with max Q|tate
(actions = ×0,9, ×1,0, ×1,1)
Adjust $\eta \leftarrow \eta * \text{action}$
Perform SGD update
$\beta_t \leftarrow \beta_t - \eta * \nabla f$
Compute new loss
Compute reward = $f_{prev} - f$
Update Q-table:
Q[state, action] ← 0,9 ∗ Q[atct‘‘] +0,1r

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